

Derivation v12: Part I Gravity & Mercury Precession

Back-Reference Audit

BLOCK-003 Bridge Map

EDC Research Program

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Abstract

One-line outcome: NO BRIDGE: Part I imports 4D gravity as [I]/[P]; BLOCK-003 remains open.

This audit examines how gravity was introduced in EDC Book Part I, specifically: (1) what was assumed/identified/postulated to obtain a 4D Newtonian/GR-like limit, and (2) how Mercury perihelion precession was computed. We produce a bridge map showing which steps can be derived from the strict 5D setup (EH+GHY+Israel) and which steps are “imported GR.”

Finding: Part I obtains gravity through a combination of dimensional analysis identification [I] and flow ansatz postulate [P]. The G formula $G = \ell_P^2 c^4 / (\sigma r_e^3)$ uses observed ℓ_P as input—explicitly acknowledged as “not an independent prediction.” The Mercury precession uses the standard GR formula with observed constants [BL]. No derivation from 5D field equations exists. BLOCK-003 (derive κ_5^2 or equivalently fix C) remains open.

1 What Part I Did: Gravity Sector

1.1 Chapter 7: Emergent Gravity from Plenum Dynamics

Part I Chapter 7 introduces gravity through the following chain:

1. **Third parameter introduced:** $\rho_{\text{Plenum}} \sim 10^{97} \text{ J/m}^3$ (Planck density scale)
2. **Physical mechanism:** Archimedean/Bjerknes attraction of pressure deficits
3. **G formula proposed:**

$$G = \frac{\ell_P^2 c^4}{\sigma r_e^3} \tag{1}$$

4. **Numerical check:** $G_{\text{calc}} = 6.71 \times 10^{-11}$ vs $G_{\text{obs}} = 6.674 \times 10^{-11}$ (1% agreement)

Critical Caveat from Chapter 7

The chapter explicitly states (line 371):

“taking ℓ_* to coincide with the measured Planck length ℓ_P makes this computation a **restatement of the chosen input scale, not an independent prediction**. A genuine prediction of G would require deriving ℓ_* (or its analogue) from EDC dynamics without inserting $\hbar$ or G as inputs.”

1.2 Chapter 8: The River Model

Chapter 8 introduces relativistic corrections via the “River Model”:

1. **Flow ansatz:** $v(r) = \sqrt{2GM/r}$ (Plenum flow toward mass)
2. **Acoustic metric:** Unruh’s result applied to Plenum $\rightarrow$ Painlevé-Gullstrand metric
3. **Schwarzschild equivalence:** Coordinate transformation gives standard Schwarzschild
4. **Mercury precession:** Standard GR formula applied

Critical Caveat from Chapter 8

The chapter explicitly states (line 278):

“once a Plenum flow ansatz ($v = \sqrt{2GM/r}$) is adopted, the Schwarzschild metric follows necessarily. The classical weak-field tests (perihelion precession, light bending, Shapiro delay) then emerge as *consistency checks*—they confirm that the mathematical structure is correct, but do not distinguish EDC from standard GR.”

2 Mercury Precession: Formula and Inputs

2.1 The Formula Used

Part I uses the **standard Einstein formula** for perihelion advance:

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1 - e^2)} \quad (2)$$

This is derived from the Schwarzschild effective potential, which itself comes from the flow ansatz.

2.2 Input Constants

Constant	Value	Source
G	$6.674 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	Observed [BL]
$M_\odot$	$1.989 \times 10^{30} \text{ kg}$	Observed [BL]
c	$2.998 \times 10^8 \text{ m/s}$	Defined [BL]
a (Mercury)	$5.79 \times 10^{10} \text{ m}$	Observed [BL]
e (Mercury)	0.2056	Observed [BL]

2.3 The Derivation Chain

Step	Tag	Content
Flow ansatz	[P]	$v(r) = \sqrt{2GM/r}$ postulated
Acoustic metric	[M]	Unruh 1981, mathematical theorem
Painlevé-Gullstrand	[D]	Follows from acoustic metric + flow
Schwarzschild	[D]	Coordinate transform from P-G
Orbit equation	[D]	Geodesic in Schwarzschild
Precession formula	[D]	Perturbation analysis
Numerical result	[BL]	Uses observed G , $M_\odot$, a , e

Result: 42.9”/century (vs observed 43.0”/century, 0.2% agreement)

3 Epistemic Classification Table

Element	Location	Tag	Justification
<i>Parameters</i>			
$\sigma \approx 1.41 \times 10^{18} \text{ J/m}^2$	Ch.6	[I]	Extracted from α, m_e, r_e
$r_e \approx 2.82 \times 10^{-15} \text{ m}$	Ch.6	[BL]	Classical electron radius
$\rho_{\text{Plenum}} \sim 10^{97} \text{ J/m}^3$	Ch.7	[P]	Postulated $\sim \rho_{\text{Planck}}$
$\ell_P \approx 1.6 \times 10^{-35} \text{ m}$	Ch.7	[BL]	Observed Planck length
<i>Gravity Mechanism</i>			
Archimedean analogy	Ch.7	[P]	Physical motivation, not derivation
$G = \ell_P^2 c^4 / (\sigma r_e^3)$	Ch.7	[I]	Dimensional identification; uses ℓ_P^{obs}
Plenum flow ansatz	Ch.8	[P]	$v = \sqrt{2GM/r}$ adopted
Schwarzschild metric	Ch.8	[D]	From acoustic metric + flow ansatz
<i>Mercury Precession</i>			
Precession formula	Ch.8	[D]	$\Delta\phi = 6\pi GM/[c^2 a(1 - e^2)]$
Numerical inputs	Ch.8	[BL]	$G, M_\odot, a, e$ observed
Result: 42.98"/century	Ch.8	[D]+[BL]	Derived formula + baseline data

Legend:

- [M] — Mathematical (pure geometry/theorem)
- [D] — Derived (follows from axioms/equations)
- [Dc] — Derived conditional (requires postulates)
- [P] — Postulate (assumed, not derived)
- [I] — Identified (definition/correspondence)
- [BL] — Baseline (calibration from measurement)

4 Bridge-to-5D Map

Part I Step	Assertion	Tag	5D Counterpart	Status	What Would Fix
ρ_{Plenum}	$\sim \rho_{\text{Planck}}$	[P]	Bulk cosmological constant Λ_5	OPEN	Derive from 5D action
σ identification	$\sigma = m_e c^2 / (\alpha r_e^2)$	[I]	Israel junction: $[K_{\mu\nu}] = -\kappa_5^2 (S_{\mu\nu} - \frac{1}{3} g_{\mu\nu} S)$	OPEN	Need κ_5^2 principle
G formula	$G = \ell_P^2 c^4 / (\sigma r_e^3)$	[I]	$G_N = \kappa_5^2 / (8\pi L)$ from KK	OPEN	Derive ℓ_P from EDC
Flow ansatz	$v = \sqrt{2GM/r}$	[P]	Weak-field limit of 5D EH	OPEN	5D perturbation theory
Schwarzschild	From acoustic metric	[D]	4D induced metric from 5D	CLOSED	(conditional on ansatz)
Precession	$\Delta\phi = 6\pi GM/[c^2 a(1 - e^2)]$	[D]	Same formula if Schwarzschild holds	CLOSED	(conditional on metric)

4.1 Gap Analysis

The bridge map reveals the following gaps between Part I and the strict 5D program:

1. **No κ_5^2 derivation:** Part I does not derive the 5D gravitational coupling from EDC field equations. The formula $G = \ell_P^2 c^4 / (\sigma r_e^3)$ uses observed ℓ_P .

2. **No EH+GHY+Israel chain:** Part I does not use the Einstein-Hilbert action, Gibbons-Hawking-York boundary term, or Israel junction conditions to obtain 4D gravity.
3. **Flow ansatz is postulated:** The key step $v = \sqrt{2GM/r}$ is adopted by analogy, not derived from 5D hydrodynamics or field equations.
4. **ρ_{Plenum} is postulated:** The Plenum density is assumed to be $\sim \rho_{\text{Planck}}$, not derived.

5 Does Part I Fix κ_5^2 or C?

5.1 The BLOCK-003 Question

BLOCK-003 asks: Can we derive G_N from EDC parameters $(\sigma, R_\xi, \rho_{\text{Plenum}})$ without circular reference to G_N^{obs} or ℓ_P^{obs} ?

The standard brane-world result is:

$$G_N = \frac{\kappa_5^2}{8\pi L} \quad (3)$$

where κ_5^2 is the 5D gravitational coupling and L is the compactification scale.

From v3-v4, we found $\kappa_5^2 = C \cdot \sigma^{-3/4}$ where C is an unfixed constant.

5.2 What Part I Provides

Part I provides the formula:

$$G = \frac{\ell_P^2 c^4}{\sigma r_e^3} \quad (4)$$

Question: Does this fix κ_5^2 or C ?

Analysis: Since $\ell_P^2 = \hbar G / c^3$, the formula becomes:

$$G = \frac{\hbar G}{c^3} \cdot \frac{c^4}{\sigma r_e^3} = \frac{\hbar G c}{\sigma r_e^3} \quad (5)$$

This is an identity (G cancels), not a constraint. It reduces to:

$$\sigma r_e^3 = \hbar c \quad (6)$$

But Part I also uses $\hbar = \sigma R_\xi^3 / c$ (from Chapter 6), giving $\sigma R_\xi^3 = \hbar c$.

These are consistent only if $r_e = R_\xi$, which is **not the case** ($r_e \sim 10^{-15}$ m, $R_\xi \sim 10^{-18}$ m).

Conclusion: The Part I formula uses ℓ_P^{obs} as input, not as output. It does not derive G or fix κ_5^2 / C .

5.3 Verdict

BLOCK-003 Status

Part I does NOT implicitly fix κ_5^2 or C.

- The G formula uses observed ℓ_P as input [BL]
- The flow ansatz is postulated [P], not derived from 5D
- Mercury precession uses observed G , $M_\odot$ [BL]
- No derivation from EH+GHY+Israel exists in Part I

BLOCK-003 remains OPEN.

6 Conclusion

6.1 Summary

Question	Answer
How does Part I get gravity?	Dimensional identification [I] + flow ansatz [P]
Is G derived from 5D?	No — uses observed ℓ_P
Is Schwarzschild derived from 5D?	No — from flow ansatz, not 5D EH
Is Mercury precession EDC-specific?	No — standard GR formula with observed inputs
Does Part I fix κ_5^2/C ?	No — BLOCK-003 remains open

6.2 One-Line Outcome

NO BRIDGE: Part I imports 4D gravity as [I]/[P]; BLOCK-003 remains open.

6.3 Implications for BLOCK-003

To close BLOCK-003, the following is needed:

1. Derive κ_5^2 from EDC action (EH+GHY+Israel)
2. Or equivalently, derive ℓ_P (or ℓ_*) from EDC dynamics
3. This requires a normalization principle for the 5D coupling
4. Part I explicitly acknowledges this gap (Chapter 7 caveat)

The Mercury precession result (42.98"/century) is a **consistency check**, not a prediction—it confirms that the adopted Schwarzschild geometry is internally consistent, but does not distinguish EDC from GR or close the derivation chain.